

Dr. Mintesnot Gizaw Terefe, PhD

Renewable Energy Feasibility Study Lead | Solar PV, Mini-Grids, PUE & Project Finance Modelling

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Renewable Feasibility, System design and installation

Professional positioning: senior renewable-energy feasibility, techno-economic modelling and climate-energy professional with PhD-level energy/climate expertise, MBA-level business training and hands-on solar PV, mini-grid, productive-use and project-finance experience.

EXECUTIVE PROFILE

Renewable energy and climate professional specialising in bankable feasibility studies for solar PV, off-grid and mini-grid systems, productive-use-of-energy (PUE) applications, water-supply and irrigation energy systems, EV charging infrastructure and climate-resilient public-service electrification. Combines PhD-level energy and climate expertise with MBA-level industrial-management training and practical project-finance modelling capability. Experienced in preparing renewable-energy feasibility studies, PVsyst/PVGIS-based energy-yield interpretation, CAPEX/OPEX build-up, LCOE, WACC, NPV, IRR, MIRR, CFADS, DSCR, sensitivity analysis, grant/debt/equity structuring and investment-grade review of renewable-energy models.

9.5+ MW	Solar-Powered PUE	Project Finance	Ethiopia & East Africa
Recent solar feasibility assignments led/reviewed	Irrigation, water supply and remote public services	NPV, IRR, DSCR, CFADS, WACC, LCOE and sensitivity	Mini-grid, energy access, EV charging and climate-energy context

CORE RENEWABLE FEASIBILITY COMPETENCIES

Competency Area	Relevant Capability
Solar PV & Mini-Grid Feasibility	Site screening, solar-resource review, PV/BESS architecture, load-demand assessment, productive-use integration, service reliability and lifecycle assumptions.
Financial and Economic Modelling	LCOE, NPV, IRR, MIRR, WACC, CFADS, DSCR, LLCR/PLCR concepts, debt schedules, grant/equity/debt structures, sensitivity and scenario analysis.
PUE Business Modelling	Solar-powered irrigation, water pumping, water-supply systems, agro-processing, remote public-service electrification and local revenue-use-case design.
PVsyst/PVGIS and Technical Review	Interpretation of PVsyst simulation outputs, PVGIS resource data, energy-yield assumptions, missing/excess energy, battery implications and bankability gaps.
Climate, ESG and Safeguards	EIA/EMS background, climate-risk framing, environmental aspects and impacts, carbon/accounting awareness and donor-style sustainability reporting.
Stakeholder and Training Delivery	Preparation of investor-ready reports, PowerPoint decks, training materials and decision-support tools for ministries, donor programmes and project teams.

SELECTED RENEWABLE-ENERGY FEASIBILITY AND MODELLING EXPERIENCE

Beles Engineering PLC — Sustainable Energy-Powered PUE Business Feasibility Study Lead

- Led renewable-energy-powered productive-use feasibility work, combining technical sizing logic, demand assessment, CAPEX/OPEX evaluation, implementation-risk review and business-case interpretation for solar-powered irrigation, water supply and remote public-service applications.

Project / Capacity	Location / Context	Feasibility Contribution
2 MW solar-powered irrigation system	Wubana Farm, Afar Regional State, Northeastern Ethiopia	Assessed solar-powered irrigation feasibility, energy demand, pumping-service concept, productive-use value creation and practical implementation considerations.
3.5 MW solar-powered multi-town water-supply project	Zone 1, Afar Regional State	Reviewed multi-town water-supply energy requirements, service reliability needs, solar PV sizing implications, operating-cost drivers and sustainability considerations.
2.5 MW solar-powered water-supply system	Pino, Kutogn, Luck-dog, Akula, Alela-2, Ebarow, Abol (G. Zuria), Kuerliye and Camp, Gambela, Ethiopia	Prepared feasibility inputs for multi-location public-service water supply using renewable energy, with attention to demand aggregation, remote-site service continuity and project prioritisation.
1.5 MW off-grid mini-grid system	Remote public-service applications	Conducted feasibility for off-grid mini-grid deployment serving remote public facilities, including preliminary load-service concept, battery/backup implications and business-model considerations.

Sodo Solar Power Plant / Standalone Mini-Grid — PVsyst Financial Model validation and Bankability Review

- Reviewed and replicated the PVsyst financial model for a standalone solar PV system of approximately 4,235 kWp with useful solar energy of about 5.898 GWh/year and major battery-storage requirements.
- Built editable, formula-linked financial analysis covering PVsyst-reported NPV, IRR, ROI, payback, energy-cost proxy, debt schedule, CFADS, DSCR and project-versus-equity return interpretation.
- Identified key feasibility flags including implicit valuation date, equity-return labelling, 15-year debt versus 25-year operating life, 10-year battery life versus 20-year depreciation, incomplete replacement treatment, grant dependency, concessional-debt dependency and limited lender-grade debt structuring.
- Prepared investment-grade correction tools: assumption register, sensitivity analysis, valuation-date impact analysis, debt-amortization review and comprehensive flag register for use by the actual modelling team.

Off-Grid Solar EV Fast-Charging Hubs — MILP-Based Feasibility and Business-Case Research

- Developed a service-constrained optimisation framework for off-grid solar PV-battery EV fast-charging hubs along the Addis Ababa–Hawassa corridor, considering corridor demand, charger sizing, PV/BESS capacity and reliability constraints.
- Integrated PVGIS solar-resource inputs, EV charging-demand synthesis, battery selection, DC fast-charger configurations, CAPEX/OPEX assumptions, WACC, NPV, DSCR/CFADS logic and sensitivity scenarios for investor-style interpretation.
- Addressed private cars, minibuses, taxis and fleet demand profiles with practical service constraints, dwell-time considerations, reliability limits and battery-storage dispatch logic.

Mini-Grid, Productive Use of Energy and Renewable Business-Modelling Support

- Prepared and delivered technical/business training materials on productive use of energy, mini-grid delivery models, business modelling, technical-economic integration of PUE and mini-grids, financing sources and financial modelling.
- Supported energy-access and mini-grid discussions linked to Power Ethiopia, UNDP and Ministry of Water and Energy training contexts, including tariff, grant, grid-arrival, repowering/decommissioning and productive-load considerations.
- Prepared renewable-energy portfolio and website materials for waste-to-energy/biomass experience, including waste-flow, life-cycle and impact framing relevant to feasibility-stage renewable project development.

PROFESSIONAL EXPERIENCE

Freelance Senior Consultant – Power Ethiopia / Energy Access and Productive Use Support | 2023–Present

- Provides renewable-energy, productive-use, mini-grid and EV-charging advisory support, including feasibility framing, training, project-document preparation and business-model interpretation.
- Supports stakeholder-facing outputs for energy-access programmes, including PowerPoint presentations, decision tools, project portfolios and financial-modelling training materials.

Assistant Professor, Sustainable Energy Technology, Department of Environmental Engineering, AASTU | Oct 2024–Present

- Teaches, supervises and develops academic materials in sustainable energy, environmental systems and renewable-energy applications.
- Links academic research with practical energy-transition, feasibility-analysis and climate-resilient infrastructure needs in Ethiopia.

Adjunct Assistant Professor, Addis Ababa University – Center for Environmental Science | Sept 2025–Present

- Supports advanced teaching and academic engagement in environmental science, climate-energy linkages and sustainability topics.

Freelance ESG Manager / Carbon Accounting and Audit Support – Explore Pharmaceutical Company | Nov 2025–Present

- Supports ESG, carbon-accounting and audit-readiness activities, strengthening the environmental and sustainability dimension of project-feasibility work.

Lecturer / Researcher and Academic Leader, Addis Ababa Science and Technology University | Jul 2012–Oct 2024

- Served as lecturer/researcher and held leadership roles including Dean and Directorate Director for Continuing and Distance Education.
- Developed teaching, research, stakeholder-management and institutional-leadership capacity applicable to multidisciplinary feasibility-study assignments.

EDUCATION AND PROFESSIONAL QUALIFICATIONS

Qualification	Institution / Field	Relevance to Renewable Feasibility
PhD – Energy & Climate	Awarded Oct 2024; University of Padova / Addis Ababa University pathway referenced	Advanced energy-climate systems analysis, research design, modelling discipline and climate-resilient project framing.
MSc / Executive Postgraduate – Strategic Energy Planning and Industrial Ecology	Politecnico di Milano	Energy-system planning, industrial ecology, life-cycle thinking and strategic renewable-energy planning.
MBA – Industrial Management	Addis Ababa Science and Technology University	Business feasibility, investment logic, management, operations and project-finance communication.
Certified Senior Consultant – Environmental Impact Assessment	MoEF Ethiopia	Environmental permitting, safeguards, compliance and impact-assessment capability for infrastructure projects.
Industrial Energy Audit Certification and RE Workshops	Professional energy-audit and ICTP/UNIDO renewable-energy training	Energy efficiency, technology assessment, renewable-energy transition and practical audit-based project evaluation.

TECHNICAL TOOLS AND METHODS

Area	Tools / Methods
Renewable technical analysis	Solar PV, BESS, off-grid and hybrid mini-grids, water pumping, irrigation loads, EV charging loads, biomass/waste-to-energy concepts.
Software and data	PVsyst report interpretation, PVGIS resource inputs, Excel financial modelling, Python/PuLP optimisation, scenario tables and graphical dashboards.
Project-finance metrics	CAPEX, OPEX, tariff, LCOE, WACC, NPV, IRR, MIRR, payback, CFADS, DSCR, debt amortisation, grant/equity/debt funding mix and sensitivity analysis.
Feasibility deliverables	Technical-economic feasibility reports, assumption registers, risk/flag registers, investment presentations, stakeholder memos, model-review notes and correction roadmaps.

SELECTED VALUE PROPOSITION FOR SIMILAR ASSIGNMENTS

- Can lead renewable feasibility studies from problem definition through site/resource assessment, load estimation, technology concept, CAPEX/OPEX build-up, financial modelling, sensitivity analysis and implementation-risk interpretation.
- Can translate technical simulation outputs into project-finance language: project NPV/IRR, equity NPV/IRR, CFADS, DSCR, debt schedule, grant dependency, replacement CAPEX, tariff affordability and bankability flags.
- Can work across technical, financial, environmental and stakeholder dimensions, which is especially valuable for donor-funded, government-supported and private-sector renewable projects in Ethiopia and similar contexts.
- Can prepare clear decision documents for project team leaders, investors, banks, donors, government agencies and technical reviewers.

References available upon request.